



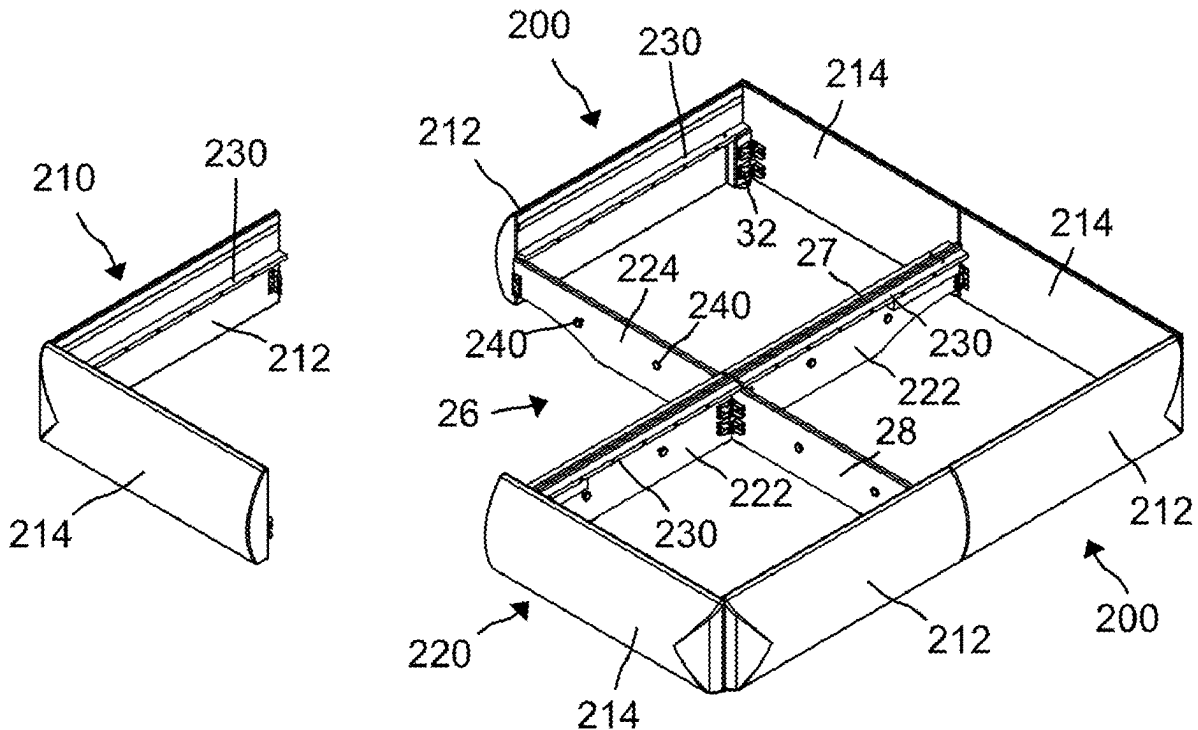
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(19) **United States**(12) **Patent Application Publication**
AUBÉ et al.(10) **Pub. No.: US 2025/0280963 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **MODULAR BED FRAME ASSEMBLY****Publication Classification**(71) Applicant: **COZEY INC.**, Mont-Royal (CA)(51) **Int. Cl.***A47C 19/00* (2006.01)*A47C 19/02* (2006.01)(72) Inventors: **Frédéric AUBÉ**, Mont-Royal (Québec)
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(2013.01); *A47C 19/027* (2013.01)(73) Assignee: **COZEY INC.**, Mont-Royal, QC (CA)(21) Appl. No.: **19/071,186**(22) Filed: **Mar. 5, 2025****Related U.S. Application Data**(60) Provisional application No. 63/561,481, filed on Mar.
5, 2024.

(57)

ABSTRACT

A modular bed frame assembly is provided. The modular bed frame assembly includes a plurality of bed frame modules adapted to be assembled in an adjacent configuration to form the modular bed frame, each of the bed frame modules being configured to support at least a portion of a bed mattress and comprising a first subassembly and a second subassembly configured to be assembled to form the respective bed frame module, each of the first and second subassemblies being configurable between a storage configuration and an assembled configuration.



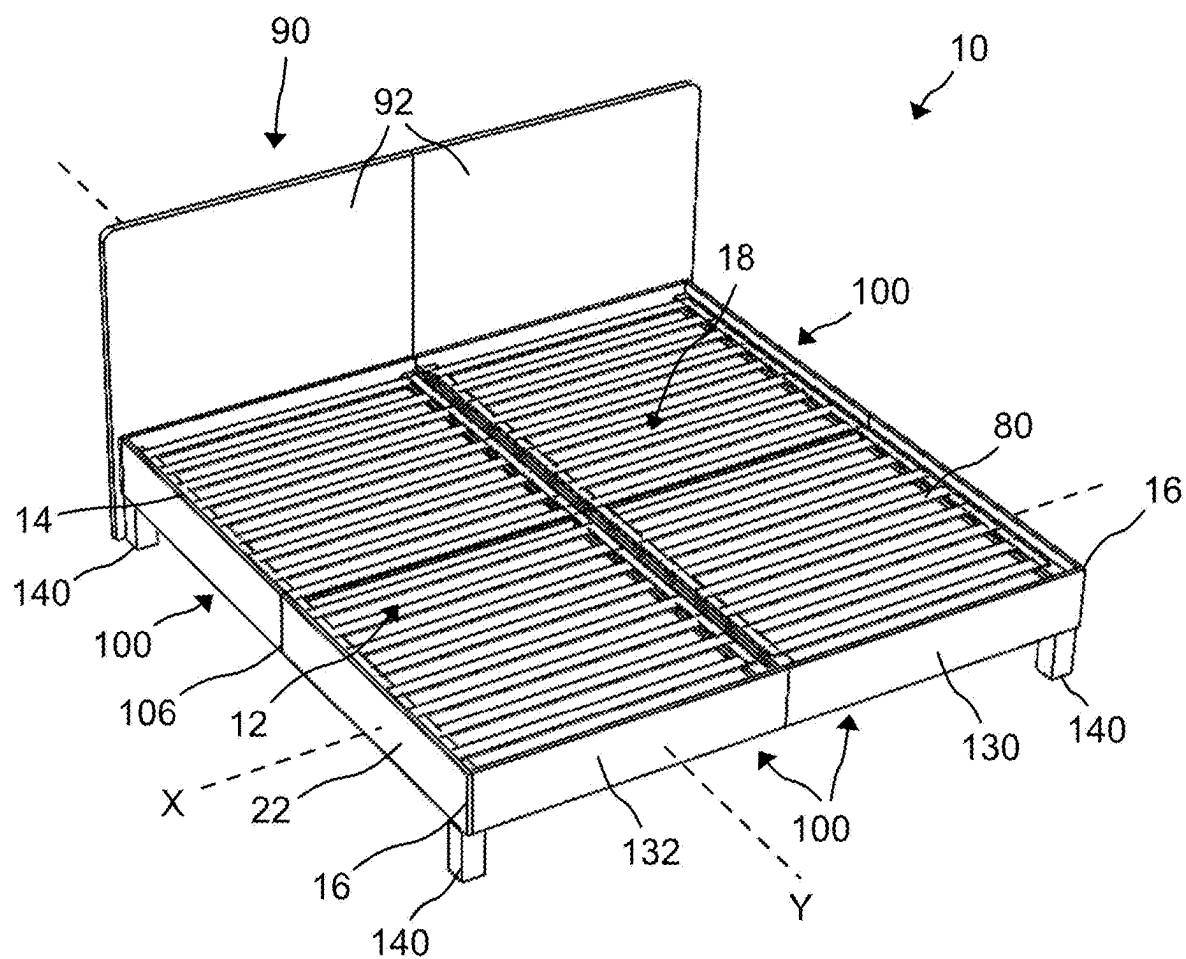
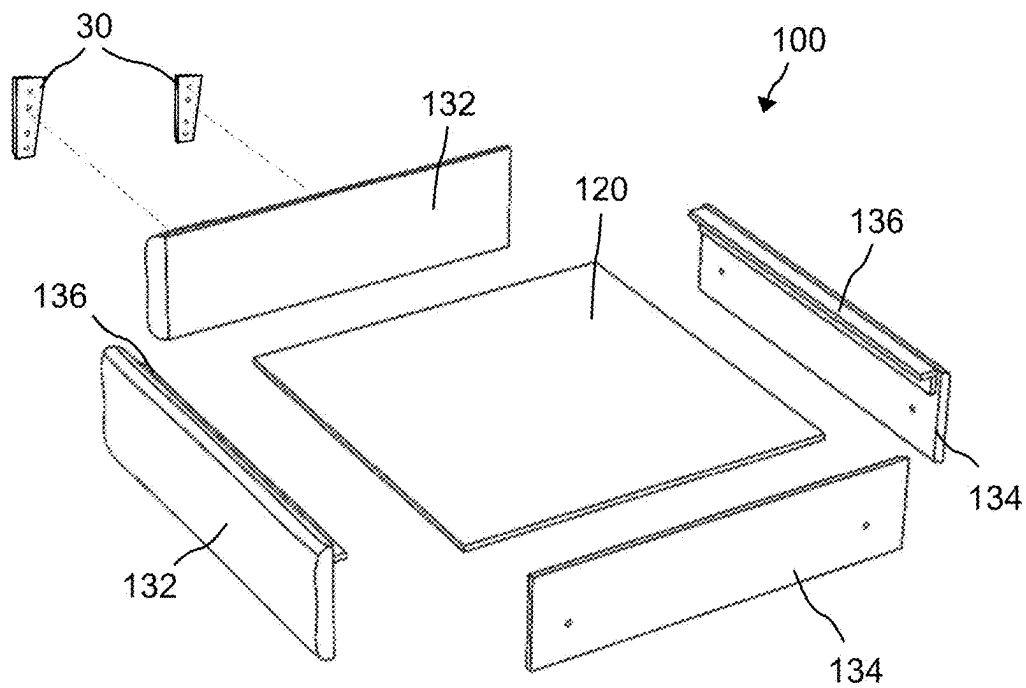
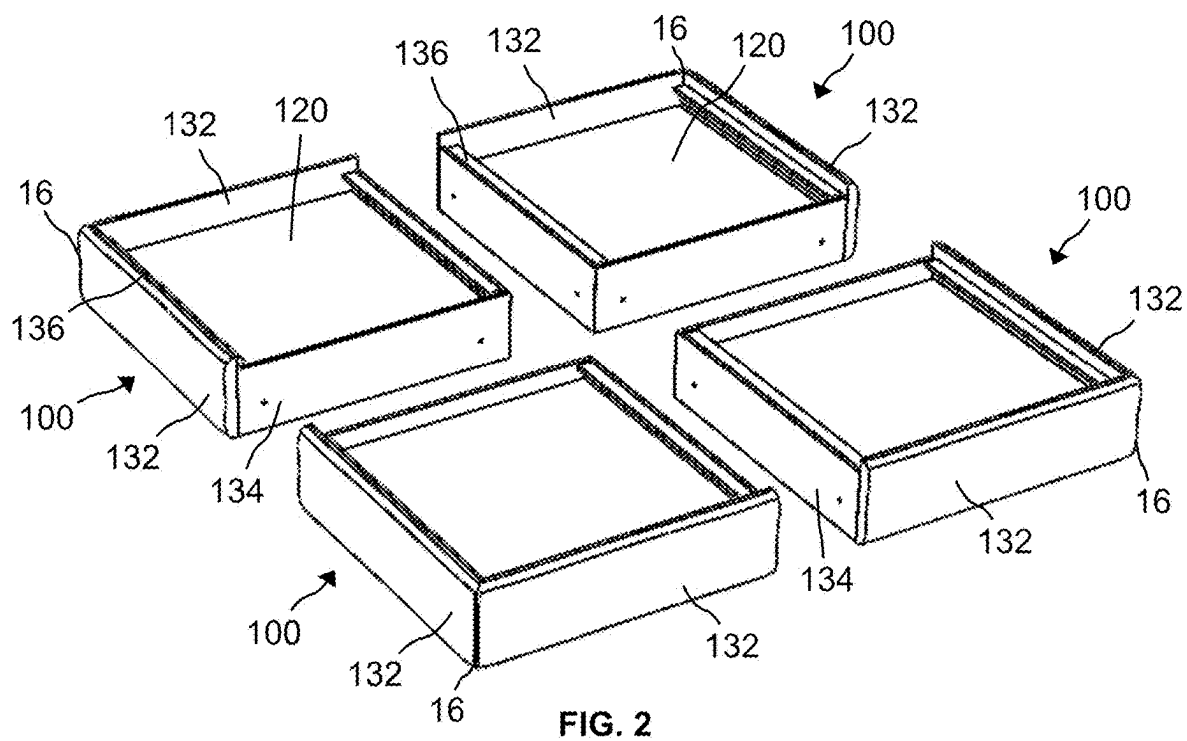


FIG. 1



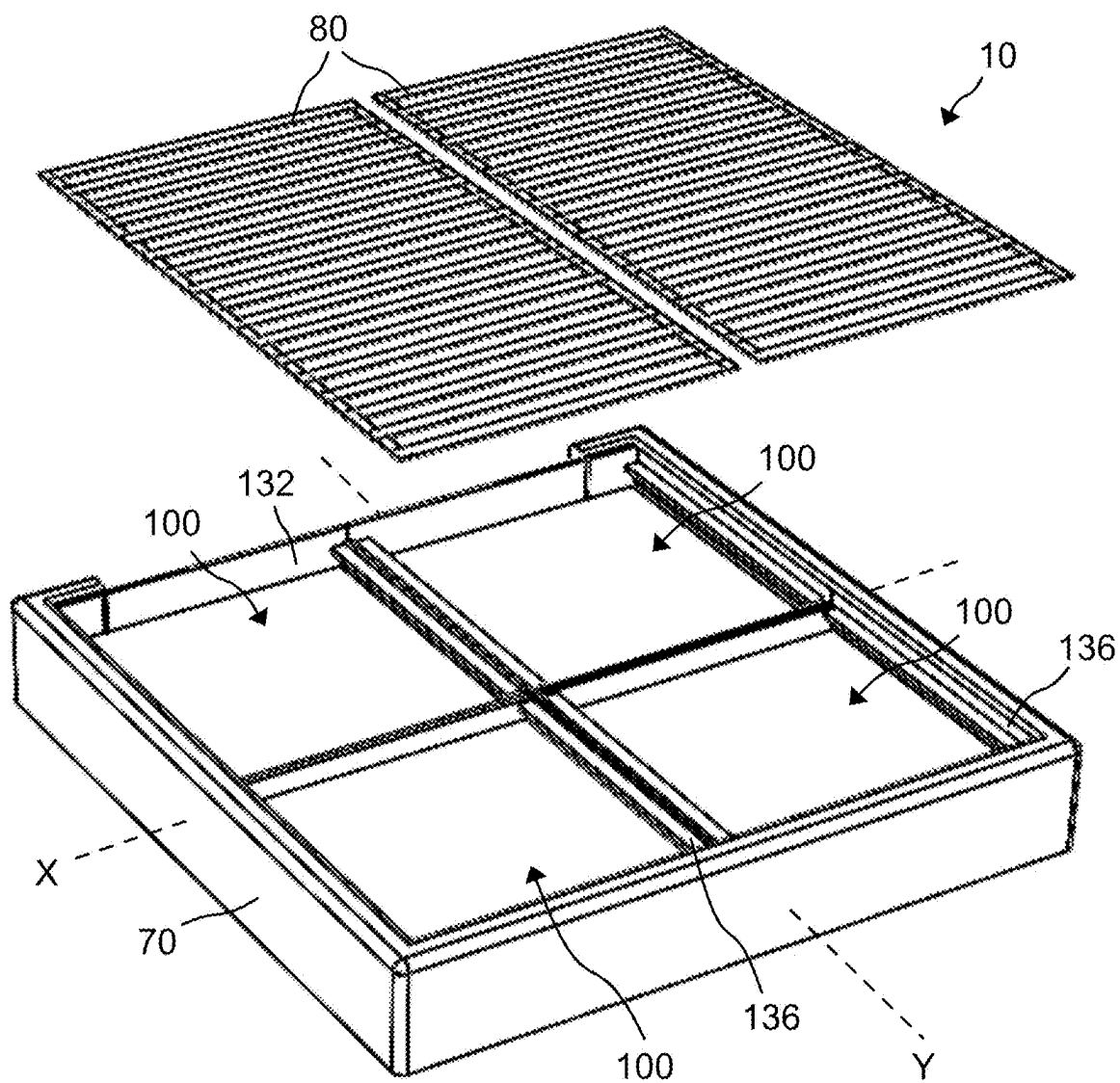


FIG. 4

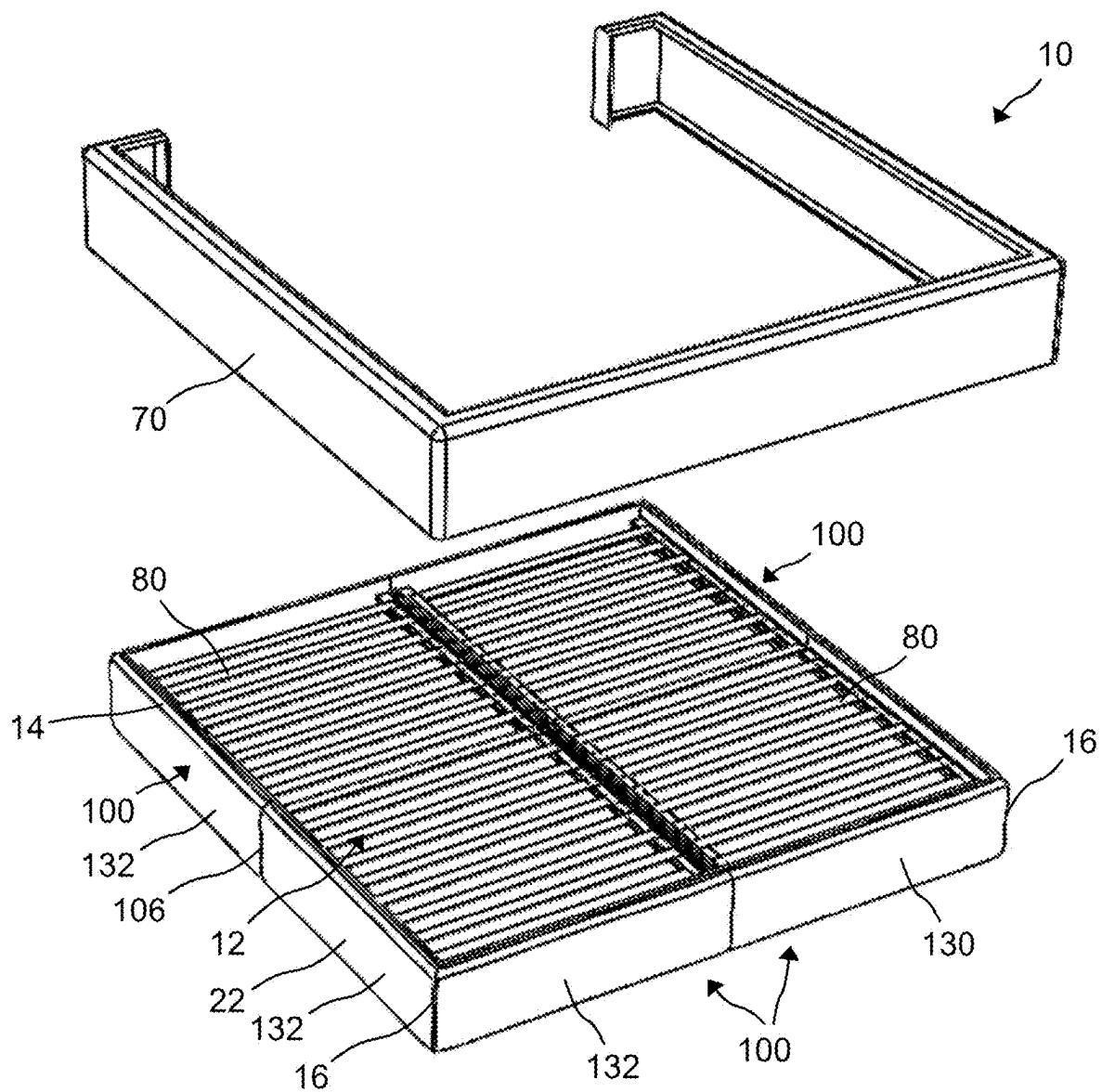


FIG. 5

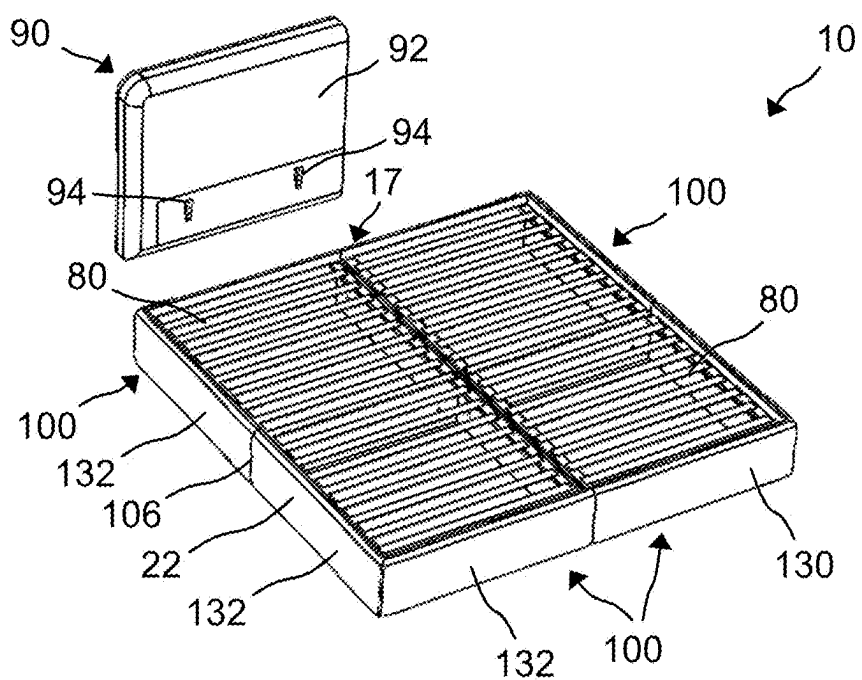


FIG. 6

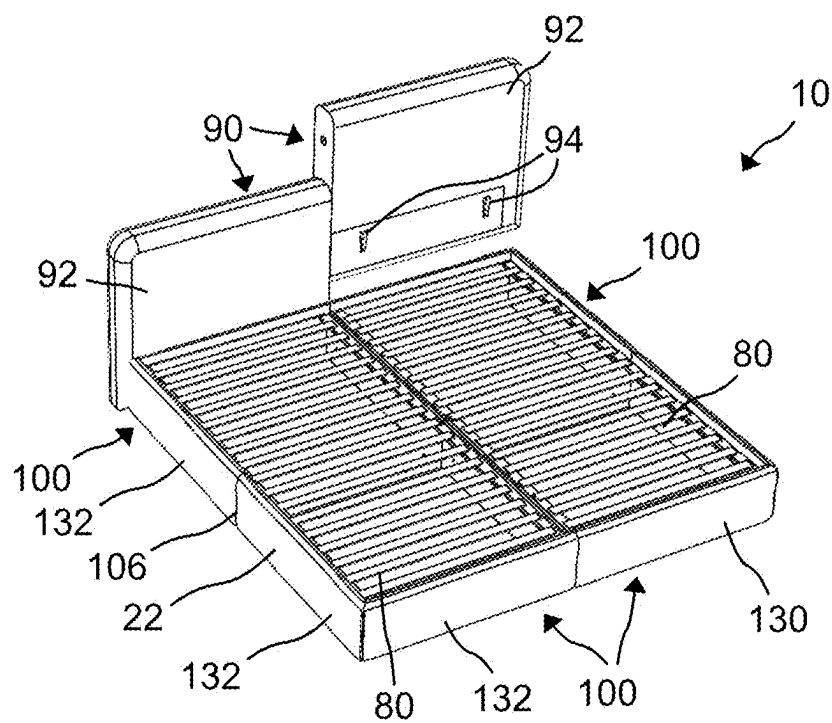


FIG. 7

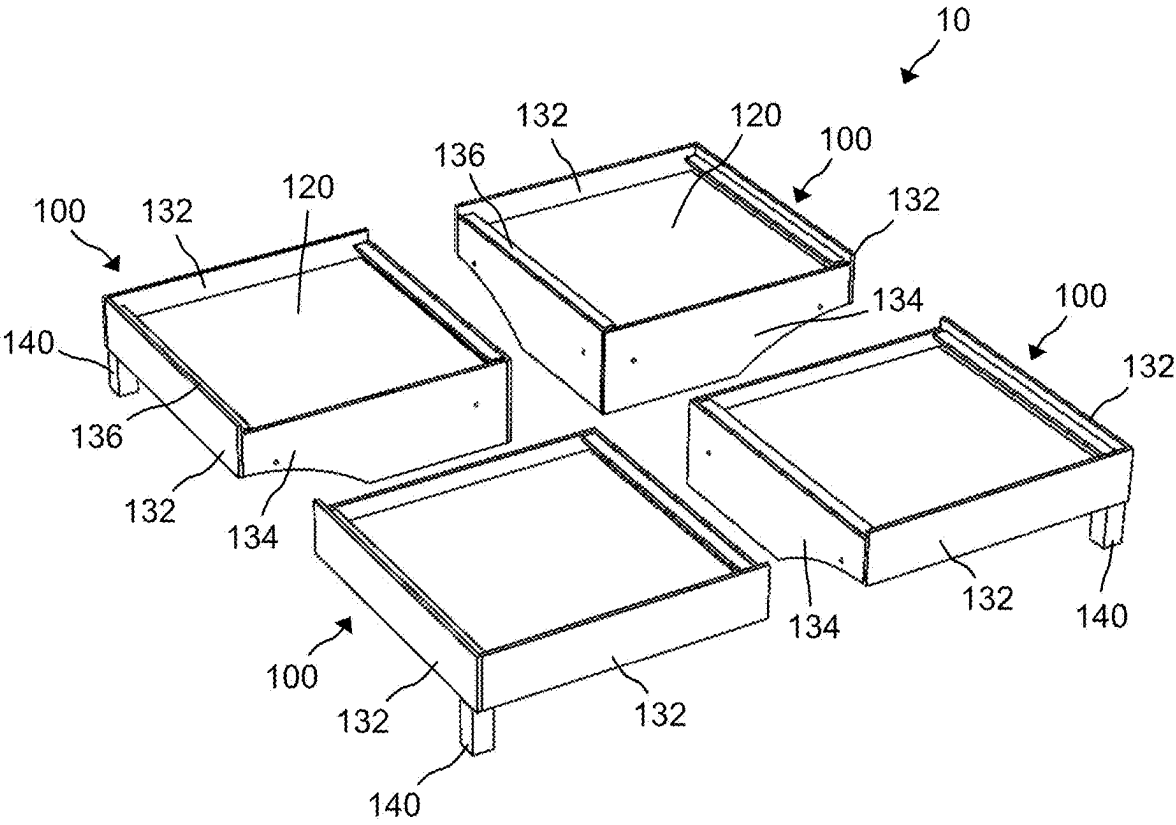


FIG. 8

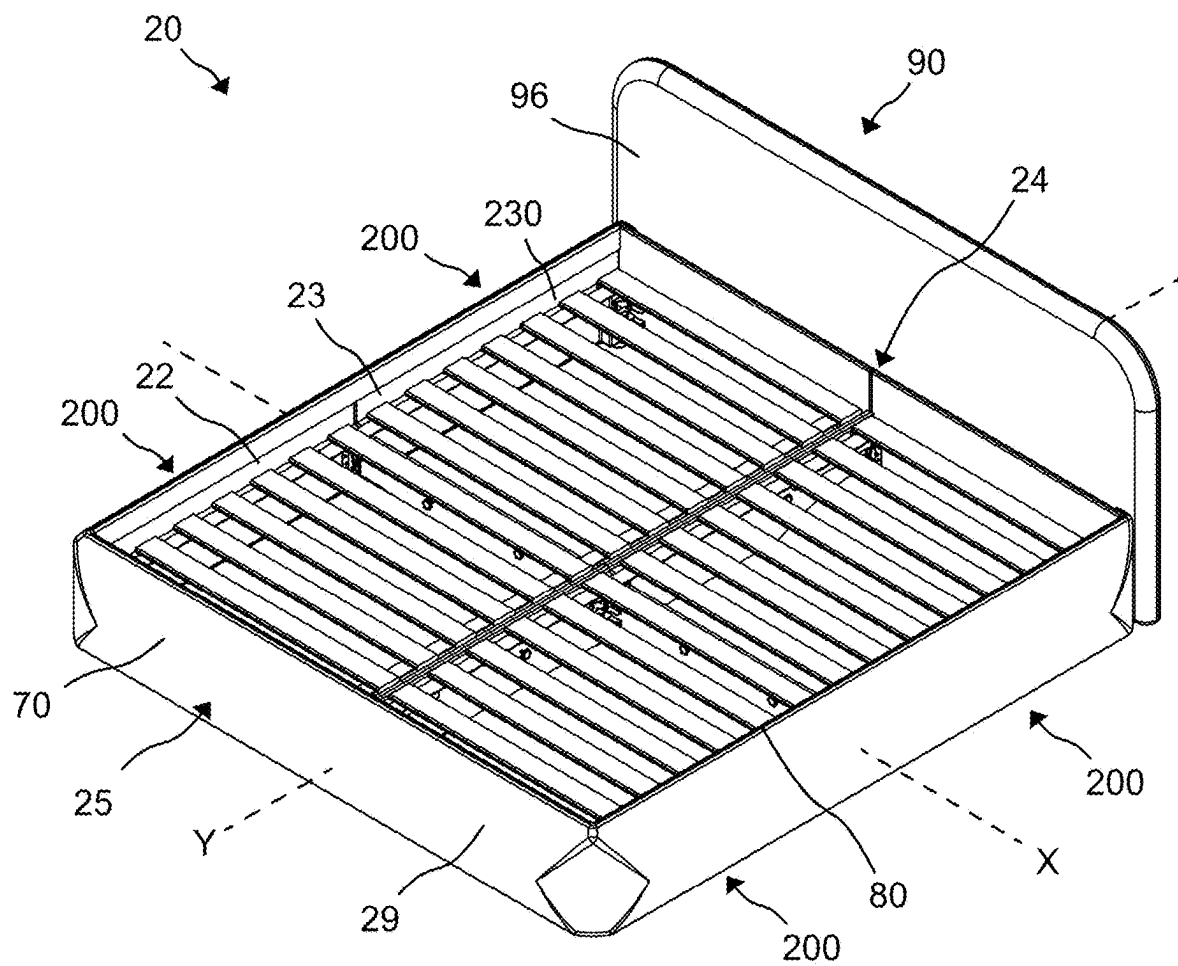


FIG. 9

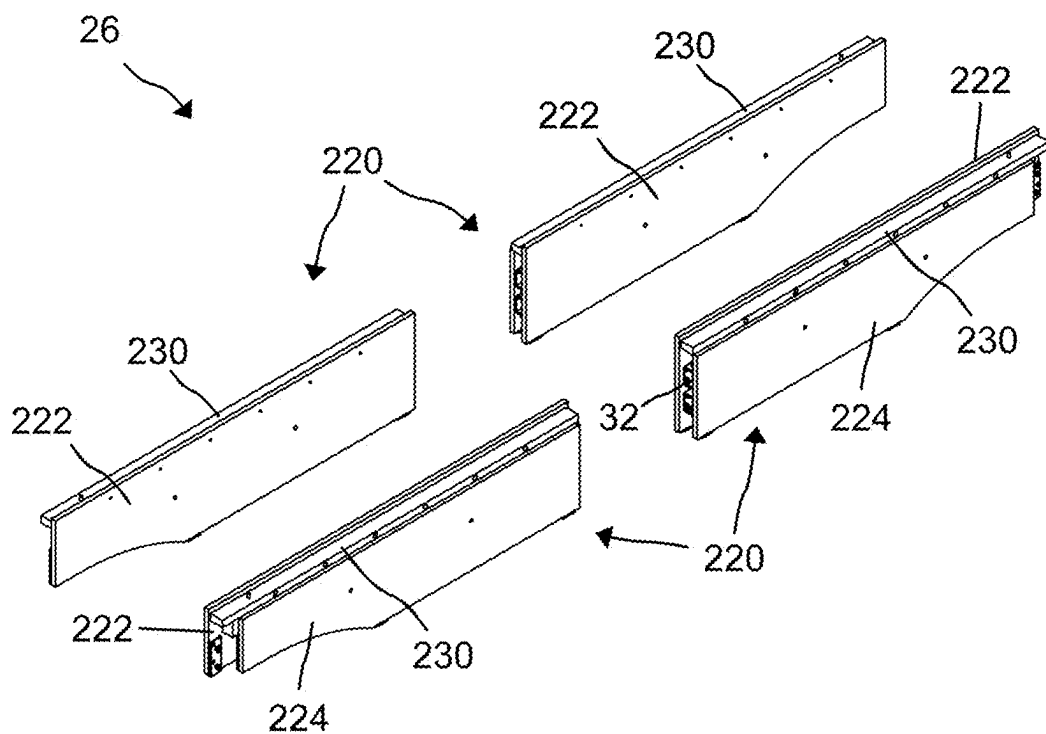


FIG. 10

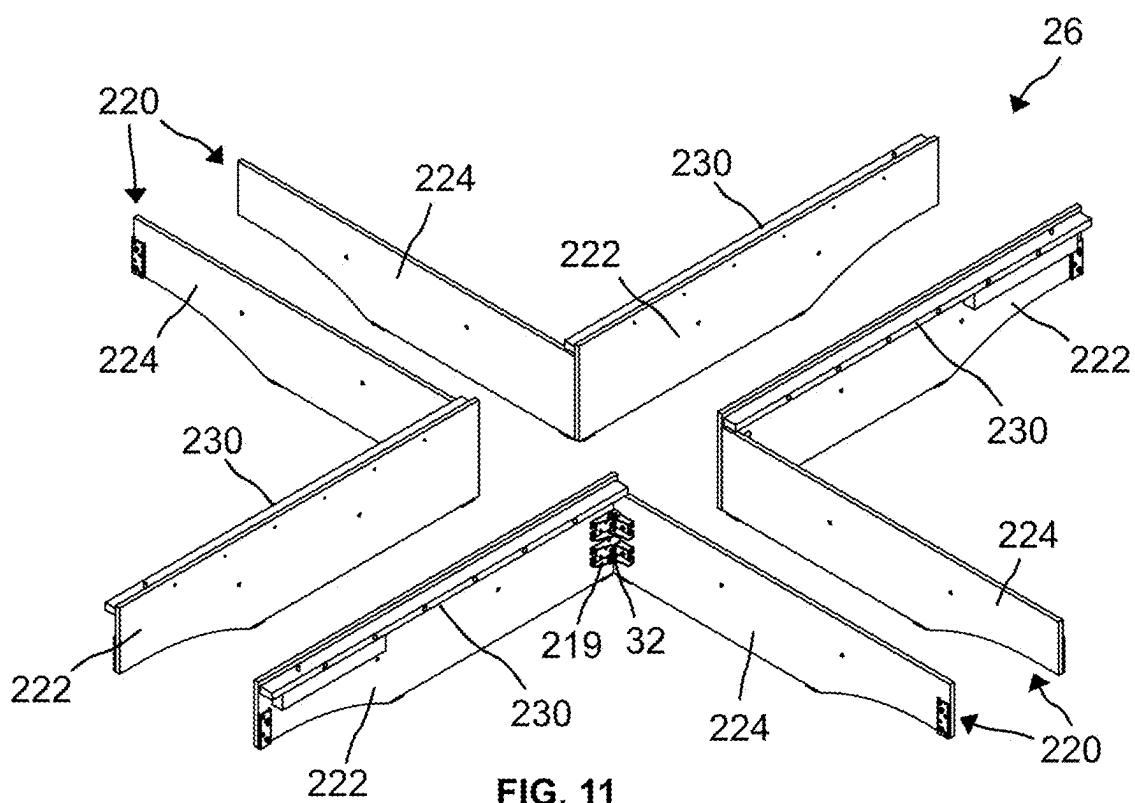


FIG. 11

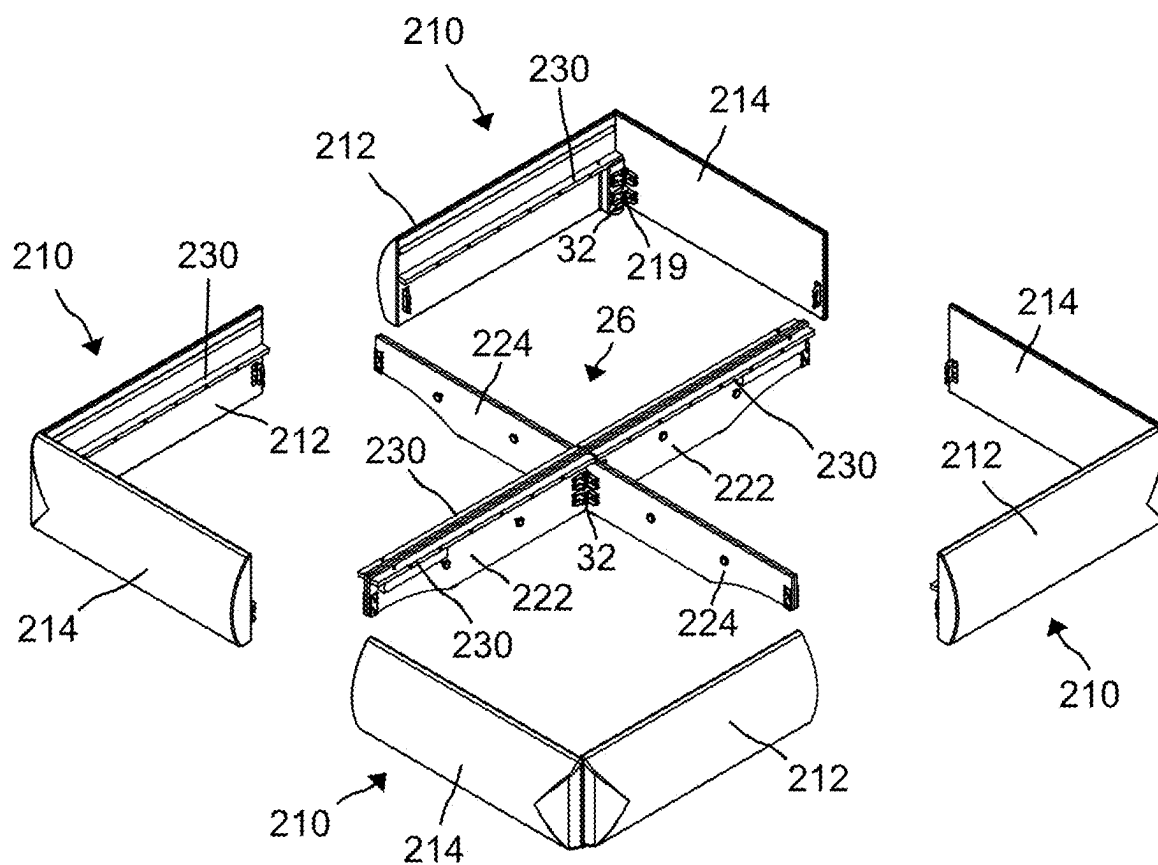


FIG. 12

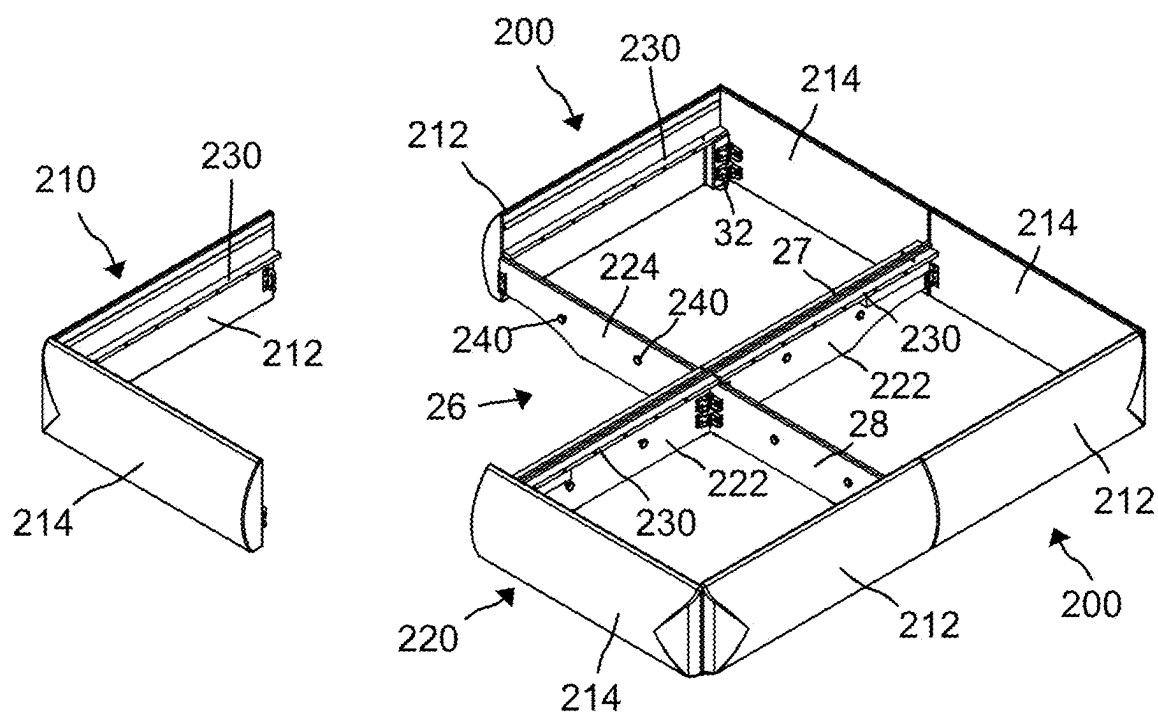


FIG. 13

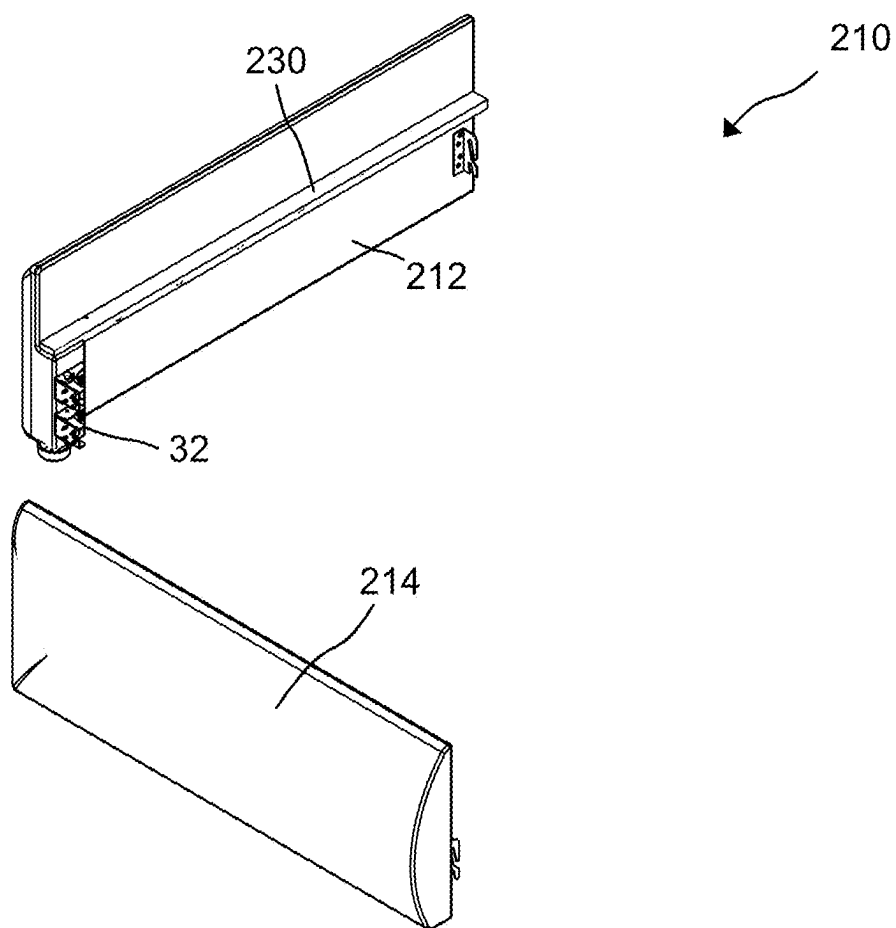


FIG. 14

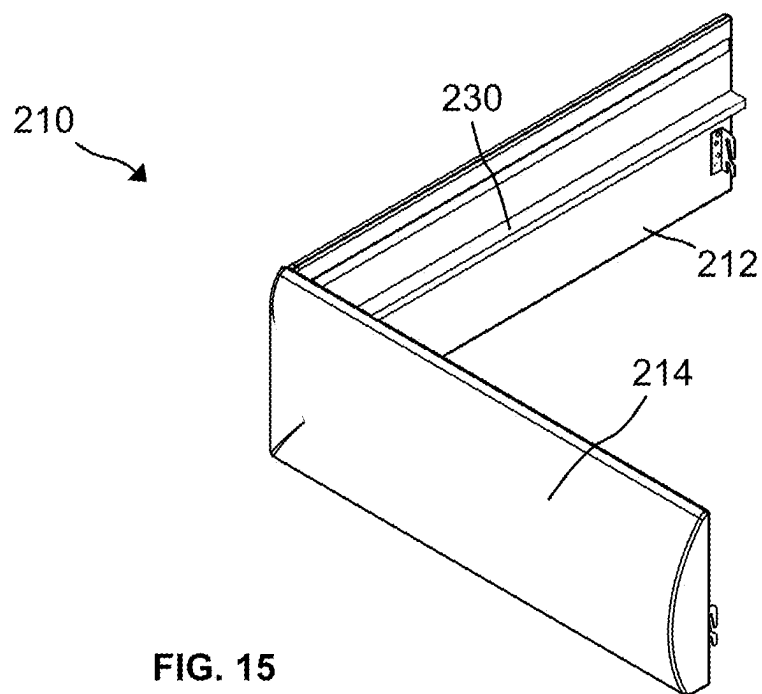


FIG. 15

MODULAR BED FRAME ASSEMBLY

RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Patent Application Ser. No. 63/561,481, filed Mar. 5, 2024, and entitled MODULAR BED FRAME ASSEMBLY, the contents of which is incorporated in its entirety herein.

TECHNICAL FIELD

[0002] The technical field generally relates to furniture, and more specifically to modular bed frame assemblies.

BACKGROUND

[0003] Traditional bed frames suffer from a number of disadvantages including the difficulty of transporting the bed frame during delivery thereof and handling of bulky structural components during assembly of the bed frame. Accordingly, existing bed frames are not normally configured to be assembled by a single person and even if they are, the task can be a cumbersome one. Moreover, the bulky structural components render traditional bed frames difficult to displace to a different space.

[0004] Finally, existing bed frames typically require the use of additional hardware and tools which may make their assembly relatively complex and/or time-consuming.

[0005] In view of the above, there is a need for a bed frame which would be able to overcome or at least minimize some of the above-discussed prior art concerns.

SUMMARY

[0006] According to one aspect, there is provided a modular bed frame assembly comprising a plurality of bed frame modules, each of the bed frame modules including an inner subassembly and a peripheral subassembly configured to be secured together to form the corresponding bed frame module, wherein the inner and peripheral subassemblies are configurable between a storage configuration with panels of the corresponding one of the inner and peripheral subassemblies extending parallel to one another, and an assembly configuration wherein the panels of the corresponding one of the inner and peripheral subassemblies extend substantially normal to one another.

[0007] According to another aspect, there is provided a modular bed frame assembly comprising a plurality of bed frame modules adapted to be assembled in an adjacent configuration to form the modular bed frame assembly, each of the bed frame modules defining at least a portion of a mattress receiving surface and comprising a first subassembly and a second subassembly configured to be assembled to form the respective bed frame module, each of the first and second subassemblies being configurable between a storage configuration and an assembled configuration.

[0008] In certain embodiments, each of the first and second subassemblies comprises first and second panels.

[0009] In certain embodiments, when the modular bed frame assembly is assembled, each of the first and second panels of the first subassembly defines a portion of a peripheral wall of the modular bed frame assembly.

[0010] In certain embodiments, the first panel of the first subassembly comprises a slat-supporting rail.

[0011] In certain embodiments, when the modular bed frame assembly is assembled, each of the first and second

panels of the second subassembly defines a portion of a central frame portion of the modular bed frame assembly.

[0012] In certain embodiments, when the modular bed frame assembly is assembled, the first panel of the second subassembly defines a portion of a center support beam of the central frame portion.

[0013] In certain embodiments, the first panel of the second subassembly includes a slat-supporting rail.

[0014] In certain embodiments, when at least one of the first and second subassemblies is in the storage configuration, the respective first and second panels extend substantially parallel to one another.

[0015] In certain embodiments, when at least one of the first and second subassemblies is in the assembled configuration, the respective first and second panels extend at an angle to one another.

[0016] In certain embodiments, when at least one of the first and second subassemblies is in the assembled configuration, the respective first and second panels are disposed orthogonally to one another.

[0017] In certain embodiments, the first panel is pivotably secured to the second panel, at least one of the first and second subassemblies including a locking hinge configured to retain the respective first panel at a given angle relative to the respective second panel when the corresponding one of the first and second subassemblies is in the assembled configuration.

[0018] In certain embodiments, the first subassembly includes an upholstered external surface.

[0019] In certain embodiments, the modular bed frame assembly further comprises a modular headboard, the modular headboard including a plurality of headboard subassemblies, each of the plurality of headboard subassemblies being securable to a corresponding one of the plurality of bed frame modules.

[0020] In certain embodiments, the modular headboard further includes a headboard cover sized to conceal the plurality of headboard subassemblies when the modular bed frame assembly is assembled.

[0021] In certain embodiments, each of the plurality of bed frame modules is disposed sequentially with a first adjacent one of the bed frame modules along a transverse direction of the modular bed frame assembly and disposed sequentially with a second adjacent one of the plurality of bed frame modules along a longitudinal direction of the modular bed frame assembly.

[0022] According to another aspect, there is provided a modular bed frame assembly comprising a plurality of central subassemblies adapted to be assembled to form a central frame of the modular bed frame assembly, and a plurality of peripheral subassemblies, each of the peripheral subassemblies being configured to be secured to the central frame of the modular bed frame assembly to define at least a portion of a peripheral wall of the modular bed frame assembly, wherein each of the central and peripheral subassemblies is configurable between a storage configuration and an assembled configuration.

[0023] In certain embodiments, each of the central and peripheral subassemblies comprises first and second panels.

[0024] In certain embodiments, the first panel is pivotably secured to the second panel.

[0025] In certain embodiments, when the corresponding one of the central and peripheral subassemblies is in the

storage configuration, the first and second panels extend substantially parallel to one another.

[0026] In certain embodiments, when the corresponding one of the central and peripheral subassemblies is in the assembled configuration, the first and second panels extend at an angle to one another.

[0027] According to another aspect, there is provided a method of assembling a modular bed frame assembly, the method comprising assembling a first bed frame module and a second bed frame module, wherein assembling the first and second bed frame modules comprises transitioning a first subassembly and a second subassembly of the corresponding first and second bed frame modules from a storage configuration to an assembled configuration, and assembling the first and second bed frame modules in an adjacent configuration to form the modular bed frame assembly.

[0028] In certain embodiments, transitioning the first subassembly from the storage configuration to the assembled configuration comprises pivoting a first panel of the first subassembly from a position parallel to a second panel of the first subassembly, to a position at an angle to the second panel of the first subassembly.

[0029] In certain embodiments, assembling the first bed frame module comprises positioning the first and second panels of the first subassembly to define a peripheral wall of the modular bed frame assembly when the modular bed frame assembly is assembled.

[0030] In certain embodiments, assembling the first bed frame module further comprises positioning a first panel of the second subassembly and a second panel of the second subassembly to define a central frame portion of the modular bed frame assembly when the modular bed frame assembly is assembled.

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. 1 is a top, front perspective view of a modular bed frame assembly, in accordance with an embodiment.

[0032] FIG. 2 is a top, front perspective view of bed frame modules of the modular sofa bed assembly illustrated in FIG. 1.

[0033] FIG. 3 is an exploded top, front perspective view of a bed frame module of the modular sofa bed assembly illustrated in FIG. 1.

[0034] FIG. 4 is a partially exploded top, front perspective view of the modular sofa bed assembly illustrated in FIG. 1.

[0035] FIG. 5 is a partially exploded top, front perspective view of the modular sofa bed assembly illustrated in FIG. 1.

[0036] FIG. 6 is a partially exploded top, front perspective view of the modular bed frame assembly illustrated in FIG. 1, shown with a first headboard module.

[0037] FIG. 7 is a partially exploded top, front perspective view of the modular bed frame assembly illustrated in FIG. 1, shown with first and second headboard modules.

[0038] FIG. 8 is a top, front perspective view of bed frame modules of the modular sofa bed assembly illustrated in FIG. 1, in accordance with another embodiment.

[0039] FIG. 9 is a top, front perspective view of a modular bed frame assembly, in accordance with another embodiment.

[0040] FIG. 10 is a top, front perspective view of subassemblies of the modular sofa bed assembly illustrated in FIG. 9, the subassemblies being shown in a storage configuration.

[0041] FIG. 11 is a top, front perspective view of the subassemblies of the bed frame modules of FIG. 10, shown in an assembled configuration.

[0042] FIG. 12 is an exploded top, front perspective view of bed frame modules of the modular sofa bed assembly illustrated in FIG. 9.

[0043] FIG. 13 is a partially assembled top, front perspective view of the bed frame modules of the modular sofa bed assembly illustrated in FIG. 9.

[0044] FIG. 14 is an exploded top, front perspective view of a subassembly of a bed frame module, in accordance with an embodiment.

[0045] FIG. 15 is a top, front perspective view of the subassembly of FIG. 14.

DETAILED DESCRIPTION

[0046] It will be appreciated that, for simplicity and clarity of illustration, where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or analogous elements or steps. In addition, numerous specific details are set forth in order to provide a thorough understanding of the exemplary embodiments described herein. However, it will be understood by those of ordinary skill in the art, that the embodiments described herein may be practiced without these specific details. In other instances, well-known methods, procedures and components have not been described in detail so as not to obscure the embodiments described herein. Furthermore, this description is not to be considered as limiting the scope of the embodiments described herein in any way but rather as merely describing the implementation of the various embodiments described herein.

[0047] For the sake of simplicity and clarity, namely so as to not unduly burden the figures with several references numbers, not all figures contain references to all the components and features, and references to some components and features may be found in only one figure, and components and features of the present disclosure which are illustrated in other figures can be easily inferred therefrom. The embodiments, geometrical configurations, materials mentioned and/or dimensions shown in the figures are optional, and are given for exemplification purposes only.

[0048] Moreover, it will be appreciated that positional descriptions such as “above”, “below”, “top”, “bottom”, “forward”, “rearward” “left”, “right” and the like should, unless otherwise indicated, be taken in the context of the figures and correspond to the position and orientation in the modular table assembly and corresponding parts when being used. Positional descriptions should not be considered limiting.

[0049] Referring to FIG. 1, there is shown a modular bed frame assembly 10, in accordance with one embodiment. In this embodiment, the bed frame assembly 10 is a rectangular construction defining a substantially planar mattress receiving surface 12 configured to directly support a mattress (not shown). When assembled, the modular bed frame assembly 10 is sized to receive one or more users in a supine position. For instance, the illustrated modular bed frame assembly 10 has a length of about 202 cm and a width of about 136 cm (commonly referred to as a “Queen” size bed) however, in other embodiments, the modular bed frame assembly 10 can have any other suitable dimensions.

[0050] Referring now to FIG. 2, the modular bed frame assembly 10 includes a plurality of bed frame modules 100

that can be assembled together to form at least a portion of the modular bed frame assembly 10. In the illustrated embodiment, the bed frame assembly 10 includes four bed frame modules 100 configured in a square configuration, i.e., with each of the bed frame modules 100 having two neighboring bed frame modules 100 positioned proximate two adjacent sides thereof. In this manner, each of the bed frame modules 100 may therefore define a peripheral corner 16 of the modular bed frame assembly 10 when the bed frame assembly 10 is assembled. In the illustrated embodiment, each of the bed frame modules 100 is disposed sequentially with an adjacent one of the bed frame modules 100 along a transverse direction X of the bed frame assembly 10 (shown in FIG. 1) and disposed sequentially with another adjacent one of the bed frame modules 100 along a longitudinal direction Y of the bed frame assembly 10 (shown in FIG. 1).

[0051] It is to be understood that the expression “modular” as used herein refers to a construction of the bed frame assembly 10 using interchangeable standardized components (i.e. the bed frame modules 100) which can be assembled to form the modular bed frame assembly 10. In accordance with some embodiments, the modular bed frame assembly 10 can therefore be assembled using a plurality of bed frame modules 100 and, if necessary or desired, reassembled to replace one or more of the bed frame modules 100 with replacement bed frame modules having a like-configuration.

[0052] In accordance with certain non-limitative embodiments, each of the bed frame modules 100 includes a central base panel 120 extending substantially horizontally and a plurality of side panels 130 secured to the central base panel 120 to define a peripheral wall of the corresponding bed frame module 100. For instance, in the illustrated embodiment, the central base panel 120 defines a bottom portion of the bed frame module 100 with the side panels 130 extending substantially upwardly therefrom. It will be understood that, in other embodiments, the bed frame modules 100 can be arranged in any other suitable manner with, for instance, the central base panel 120 defining an upward portion of the bed frame module 100 with the side panels 130 extending substantially downwardly therefrom. In other embodiments still, the central base panel 120 can be omitted with the side panels 130 forming the structure of the corresponding bed frame module 100.

[0053] In certain embodiments, the side panels 130 can include two peripheral side panels 132 positioned along two adjacent sides of the corresponding bed frame module 100 to define, when the modular bed frame assembly 10 is assembled, a portion of an external peripheral wall 22 of the bed frame assembly 10. The side panels can further include two central side panels 134 positioned along the opposite two adjacent sides of the corresponding bed frame module 100. Configured in this manner, each of the central side panels 134 can extend from an adjacent one of the external peripheral walls 22 towards a central region 18 of the modular bed frame assembly 10, i.e., a region of the modular bed frame assembly 10 located inwardly from the peripheral wall 22 that is substantially concealed when the modular bed frame assembly 10 is assembled. In some embodiments, each of the side panels 130 can be secured to one or more adjacent ones of the side panels 130 and/or to the base panel 120 using one or more frame connectors 30. For instance, the frame connectors 30 can include and fastener suitable for

securing the side panels 130 including, for instance, brackets, clamps, screws, an adhesive, or the like.

[0054] In certain embodiments, the bed frame assembly 10 can further be configured to at least partially prevent a lateral displacement of the mattress when said mattress is positioned on the mattress receiving surface 12. More specifically, in accordance with certain embodiments, one or more of the peripheral side panels 132 can be sized to extend upwardly above the mattress receiving surface 12 to define a peripheral edge 14 along at least a portion of a periphery of the mattress receiving surface 12. In certain embodiments, the peripheral edge 14 can thus limit a lateral displacement of the mattress resting upon the mattress receiving surface 12.

[0055] In some embodiments, one or more of the side panels 130 can include a slat-supporting rail 136 configured to at least partially support a slat assembly 80 (shown in FIG. 4). In the embodiment illustrated in FIG. 3, each of opposing ones of a peripheral side panel 132 and a central side panel 134 includes a slat-supporting rail 136 to cooperatively support a portion of the slat assembly 80. In the illustrated embodiment, the peripheral side panel 132 and the inner side panel 134 can be disposed opposite one another and extend in the longitudinal direction Y of the bed frame assembly 10 to support opposite sides of the slat assembly 80. It is understood that, in other embodiments, any other suitable ones of the side panels 130 can include the slat-supporting rail 136 for supporting the slat assembly 80.

[0056] Referring to FIG. 4, in some embodiments, adjacent ones of the bed frame modules 100 of the modular bed frame assembly 10 can be configured to cooperatively support the slat assembly 80. For instance, adjacent bed frame modules 100 aligned in the longitudinal direction Y can cooperatively support a single slat assembly 80. It will be appreciated that, in this manner, a user resting on the modular bed frame assembly 10 can be supported by a single slat assembly 80. In such embodiments, the central side panel 134 extending along the transverse direction X of the modular bed frame assembly 10 can be sized to provide an unobstructed extension of the slat assembly 80 across the adjacent bed frame modules 100. More specifically, a height of the central side panel 134 extending in the transverse direction X of the modular bed frame assembly can be lower than a height of the slat-supporting rail 136.

[0057] When the bed frame modules 100 are assembled as described above, the bed frame modules 100 can be positioned in an adjacent configuration in accordance with the desired shape and configuration of the modular bed frame assembly 10. In order to assemble the modular bed frame assembly 10, the bed frame modules 100 can be secured together using the frame connectors 30 described above or using any other suitable fastener. In other embodiments, one or more of the bed frame modules 100 may be sufficiently heavy or weighted down to maintain a fixed position of the bed frame modules 100 relative to their adjacent bed frame modules 100 thereby eliminating the need for the frame connectors 30. In the illustrated embodiment, each of the bed frame modules 100 includes multiple frame connectors 30 configured to enable an assembly of the bed frame modules 100 to each other and/or to accessory components of the modular bed frame assembly 10, such as a headboard, described in greater detail below. When used for connecting two adjacent ones of the bed frame modules 100, the frame connectors 30 can be secured to the central side panels 134

or any other suitable element of the corresponding bed frame module 100. As described above, the term “frame connector” as used herein, such as the frame connectors 30, is intended to include any coupling mechanism configured to hold together adjacent ones of the bed frame modules 140 of the modular bed frame assembly 10 in a side-by-side relationship, or to secure an accessory component thereto. Individual ones of the frame connectors 30 may be referred to as hand screws, clips or hooking members.

[0058] When the bed frame assembly 10 is assembled, i.e., when adjacent ones of the bed frame modules 100 are positioned in a side-by-side relationship, a module gap or module seam 106 may be formed between the adjacent bed frame modules 100. In particular, in certain embodiments, the module seam 106 may be visible along the peripheral wall 22 of the modular bed frame assembly 10 when assembled. Referring now to FIG. 5, in some embodiments, the modular bed frame assembly 10 can further include a decorative covering 70 shaped and sized to overlay at least a portion of the peripheral wall 22 defined by the peripheral side panels 132. The decorative covering 70 may provide a protection to the bed frame modules 100 or at least partially conceal the bed frame modules 100 (including, for instance, concealing a module seam 106 formed between adjacent ones of the bed frame modules 100). In some embodiments, the decorative cover 70 can be a padded or upholstered construction comprising a textile exterior forming a hollow interior stuffed with a soft non-woven material to provide a protection to the peripheral side panels 132 of the bed frame modules 100.

[0059] Referring now to FIGS. 6 and 7, an embodiment of the bed frame assembly 10 is shown with a modular headboard 90 configured to be secured to a longitudinal upper end 17 of the modular bed frame assembly 10. The modular headboard 90 can include one or more headboard subassemblies 92 configured to be secured to one or more of the bed frame modules 100 and/or to an adjacent one of the headboard subassemblies 92. In such embodiments, the one or more bed frame modules 100 disposed proximate the longitudinal upper end 17 of the bed frame assembly 10 in the longitudinal direction Y of the bed frame assembly 10 can include one or more frame connectors 30 configured to matingly engage corresponding headboard connectors 94 of the headboard subassemblies 92. In the illustrated embodiment, the frame connectors 30 of the bed frame modules 100 and the headboard connectors 94 of the headboard subassemblies 92 are male and female sliding connectors, respectively, configured to slidably engage when the respective headboard subassembly 92 is translated downwardly for installation onto the corresponding bed frame module 100.

[0060] In certain embodiments, when the modular headboard 90 includes multiple headboard subassemblies 92, the modular headboard can further include a headboard cover 96 sized to conceal the plurality of headboard subassemblies 92 when the modular bed frame assembly 10 is assembled. More specifically, the headboard cover 96 can be configured to at least partially conceal the headboard subassemblies 92 (including, for instance, concealing a seam formed between adjacent ones of the headboard subassemblies) while maintaining access to the headboard connectors 94. In some embodiments, the headboard cover 96 can be a padded or upholstered construction comprising a textile exterior form-

ing a hollow interior stuffed with a soft non-woven material to provide a protection to the headboard subassemblies 92 of the headboard module 90.

[0061] In the illustrated embodiment, each of the bed frame modules 100 disposed proximate the longitudinal upper end 17 of the bed frame assembly 10 is configured to be assembled with a corresponding headboard subassembly 92. Alternatively, in other embodiments, the modular bed frame assembly 10 can include a single headboard module 90 sized to be coupled with multiple bed frame modules 100.

[0062] In some embodiments, one or more of the side panels 130 of the bed frame modules 100 can be configured to contact a ground surface to provide a support to the corresponding bed frame module 100. For instance, in the illustrated embodiment, each of the side panels 130 is dimensioned to abut the ground surface so as to at least partially support a weight of the bed frame module 100 as well as the mattress resting on the bed frame assembly 10 and any user(s) resting thereupon. Referring now to FIG. 8, in other embodiments, one or more of the bed frame modules 100 can include a support member 140 configured to be secured to one or more of the base panel 120 and the side panels 130. The support member 140 can extend downwardly therefrom to the ground surface to at least partially support the corresponding bed frame module 100. In such embodiments, each support member 140 can include a connection portion connectable to the base panel 120 and a support portion extending away from the connection portion for contacting the ground surface.

[0063] Referring now to FIGS. 9 to 15, there is shown a modular bed frame assembly 20, in accordance with another embodiment. The modular bed frame assembly 20 includes a plurality of bed frame modules 200 adapted to be assembled in an adjacent configuration to form the modular bed frame assembly 20. Notably, in contrast with the modular bed frame assembly 10 describe above, one or more of the bed frame modules 200 comprises collapsible subassemblies which can be configured in one of a storage configuration and an assembled configuration. It will be understood that, as used herein, the expression “subassemblies” is intended to refer to pre-constructed components that are assembled together to form a part of the bed frame module 200 and to provide structural strength and stability thereto. In this manner, the expression “subassemblies” is not intended to refer to smaller individual elements including fasteners used to connect or secure the subassemblies.

[0064] Referring to FIGS. 10 to 15, in the illustrated embodiment, each of the bed frame modules 200 includes a first subassembly 210 and a second subassembly 220 configured to be assembled to form the corresponding bed frame module 200. In the illustrated embodiment, the first subassembly 210 includes first and second panels 212, 214 which, when the modular bed frame assembly 20 is assembled, each define a portion of a peripheral wall 22 of the modular bed frame assembly 20. For instance, in the embodiment illustrated in FIG. 13, the first panel 212 of the first subassembly 210 defines a portion of a side rail 23 of the modular bed frame assembly 20 (shown in FIG. 9) extending along a longitudinal axis Y thereof. In certain embodiments, the second panel 214 of a given one of the bed frame modules 200 disposed at a longitudinal lower end 25 of the modular bed frame assembly 20 can define a portion of a footboard 29 of the modular bed frame assembly 20 extending along a transverse axis X thereof. In such embodiments, the first

subassembly 210 can thus be referred to as a “peripheral subassembly” and the first and second panels 212, 214 can be referred to as “peripheral panels”.

[0065] In certain embodiments, when the first subassembly 210 is a peripheral subassembly, the second subassembly 220 can include first and second panels 222, 224 defining a portion of a central frame 26 of the modular bed frame assembly 20. In such embodiments, the second subassembly 220 can be referred to as a central subassembly and the first and second panels 222, 224 can be referred to as central panels. In certain embodiments, the central frame 26 of the modular bed frame assembly 20 can form an internal structural component of the modular bed frame assembly 20 configured to provide structural support and stability thereto, ensuring that all the components of the modular bed frame assembly 20 are securely held in place. In this manner, the central frame 26 can act as the foundational structure of the modular bed frame assembly 20 to distribute weight and absorb stress. The expression “internal structural component” is intended to refer to a component of the modular bed frame assembly 20 that is substantially concealed when the modular bed frame assembly 20 is assembled.

[0066] In certain embodiments, the central frame 26 of the modular bed frame assembly 20 can be self-supporting, i.e., the central frame 26 can be configured to contact the ground surface and support a weight of the modular bed frame assembly 20 when assembled. To that end, one or more of the first and second panels 222, 224 can be sized to contact the ground surface beneath the central frame 26 and/or the central frame 26 can include one or more supporting members configured to at least partially support the central frame 26, similar to the support members 140 described above.

[0067] In certain embodiments, the first subassembly 210 can include a pivoting connector 32 to pivotably connect the first and second panels 212, 214. When configured in this manner, the first and second panels 212, 214 of the first subassembly 210 can be pivotally hinged to one another to enable the first subassembly 210 to transition between the storage configuration and the assembled configuration. The pivoting connector 32 can include a hinged connector or any other connector suitable for enabling a pivoting connection between the first and second panels 212, 214. Similarly, in certain embodiments, the first and second panels 222, 224 of the second subassembly 220 can be pivotally hinged to one another via a hinged connector 32 to enable the second subassembly 220 to transition between the storage configuration and the assembled configuration.

[0068] Referring now to FIG. 10, the second subassemblies 220 of multiple bed frame modules 200 are shown in a collapsed configuration. When in the collapsed configuration, the first and second panels 222, 224 are pivoted such that the first panel 222 extends substantially parallel to the second panel 224. Referring now to FIG. 11, the second subassemblies 220 of multiple bed frame modules 200 are shown in an assembled configuration. When in the assembled configuration, the respective first and second panels 222, 224 extend at an angle to one another. For instance, in the illustrated embodiment, the first and second panels 224, 224 are disposed orthogonally to one another when the corresponding second subassembly 220 is in the assembled configuration.

[0069] In certain embodiments, the first subassembly 210 can be configured similarly to the second subassembly 220. More specifically, the first and second panels 212, 214 can

be pivoted such that the first panel 212 extends substantially parallel to the second panel 214 when the first subassembly is in the storage configuration. Moreover, the first and second panels 212, 214 of the first subassembly 210 can extend at an angle to one another when the first subassembly 210 is in the assembled configuration with, in certain embodiments, the first and second panels 212, 214 of the first subassembly 210 extending orthogonally to one another when the first subassembly 210 is in the assembled configuration.

[0070] While the illustrated embodiment includes first and second subassemblies 210, 220 including a pivoting connector 32, it will be understood that, in other embodiments, the first and second subassemblies 210, 220 can be assembled using any other type of connector including, for instance, connectors which restrict a pivoting between the corresponding first and second panels. For instance, in certain embodiments, the corresponding first and second panels can be detached to adopt the storage configuration and attached to adopt the assembled configuration.

[0071] It will be appreciated that, when the bed frame modules 200 of the modular bed frame assembly 20 are disassembled and the corresponding first and second subassemblies 210, 220 are in the storage configuration, the modular bed frame assembly 20 may have a reduced volume with reduced dimensions in comparison to a traditional bed frame thus promoting efficient use of space and organization. Accordingly, the configuration of the bed frame modules 200 may facilitate transportation of the modular bed frame assembly 20 and the use of components having a reduced size may facilitate installation thereof.

[0072] In certain embodiments, each of the first and second subassemblies 210, 220 includes a pivot stopper 219 configured to limit a pivoting rotation of the corresponding first and second panels beyond a predetermined angle. For instance, in some embodiments, the pivot stopper 219 can be configured to limit a pivoting rotation of the corresponding first and second panels beyond an angle of 90°. The pivot stopper 219 can be configured as a wall member, a flexible tension member or any other component configured to limit a pivoting rotation between hinged members. For instance, in the illustrated embodiment, the pivot stopper 219 is integrated into the pivoting connector 32 provided as a locking hinge. In such embodiments, the pivoting connector 32 including the pivot stopper 219 is configured to retain the respective first panel at a given angle relative to the respective second panel when the corresponding one of the first and second subassemblies 210, 220 is in the assembled configuration.

[0073] Referring now to FIGS. 12 and 13, as specified above, the second subassemblies 220 of the bed frame modules 200 can be secured together to form the central frame 26 of the modular bed frame assembly 20. More specifically, the first panel 222 of each of the second subassemblies 220 can define a portion of a central support beam 27 of the modular bed frame assembly 20. The central support beam 27 can extend longitudinally from a longitudinal upper end 24 of the modular bed frame assembly 20 to the longitudinal lower end 25 thereof. Similarly, in certain embodiments, the second panel 224 of each of the second subassemblies 220 can define a portion of a transverse support beam 28 of the modular bed frame assembly 20. The

transverse support beam 28 can extend transversely between opposing ones of the side rails 23 of the modular bed frame assembly 20.

[0074] It will be understood that, in certain embodiments, the central frame 26 of the modular bed frame assembly 20 can help prevent mattress sagging, provide additional stability and durability, ensure an even weight distribution, and/or maintain an overall structural integrity of the modular bed frame assembly 20.

[0075] The second subassemblies 220 can be secured together using one or more frame connectors 240 or any other suitable connector. Once the rigid inner frame 220 is assembled, one or more of the peripheral assemblies 216 can be secured thereto to form the bed frame assembly 20 (as shown in FIG. 13).

[0076] In certain embodiments, one or more of the bed frame modules 200 can include a slat-supporting rail 230 configured to at least partially support the slat assembly 80. In the embodiment illustrated in FIGS. 12 and 13, each of the first panels 212, 214 of the first and second subassemblies 210, 220 includes a slat-supporting rail 230 to cooperatively support a portion of the slat assembly 80 when the bed frame assembly 20 is assembled. In certain embodiments, each of the bed frame modules 200 can be configured to receive a distinct slat assembly 80. In the illustrated embodiment, the transverse support beam 28 of the central frame 26 is sized to enable an extension of the slat assembly 80 across multiple bed frame modules 200 aligned in the longitudinal direction Y of the modular bed frame assembly 20. More specifically, a height of the second panel 224 of the second subassemblies 220 can be limited to not extend above a height of the slat-supporting rail 230. In this manner, the slat assembly 80 can extend across multiple bed frame modules 200 aligned in the longitudinal direction Y without obstruction. It will be understood that the slat assembly 80 extending across multiple bed frame modules 200 can provide better support for the mattress by more evenly distributing a weight of the mattress across the modular bed frame assembly 20, thus potentially further preventing sagging. The slat assembly 80 extending across multiple bed frame modules 200 can further improve airflow and reduce pressure points, contributing to a more comfortable and durable sleeping surface.

[0077] A method of assembling the modular bed frame assembly 20 will now be described in further detail. The method can include assembling multiple bed frame modules 200. The assembly of the bed frame modules 200 can include transitioning the first subassembly 210 and the second subassembly 220 of the respective bed frame modules 200 from a storage configuration to an assembled configuration and securing the first and second subassemblies 210, 220 together to form the corresponding bed frame module 200. The method can further include assembling the bed frame modules 200 in an adjacent configuration to form the modular bed frame assembly 20. In certain embodiments, the method can include securing the bed frame modules 200 together using one or more frame fasteners.

[0078] In certain embodiments, transitioning the first and second subassemblies 210, 220 from the storage configuration to the assembled configuration can include pivoting the first panel 212, 222 of the first and second subassemblies 210, 220 from a position parallel to the second panel 214, 224 of the first and second subassemblies 210, 220, to a position at an angle to the second panel 214, 224.

[0079] In other embodiments, a method of assembling the modular bed frame assembly 20 can include transitioning the second subassemblies 220 of multiple bed frame modules 200 from a storage configuration to an assembled configuration as described above. The method can further include securing the second subassemblies 220 together to form the central frame 26 of the modular bed frame assembly 20. In certain embodiments, the method can further include transitioning the first subassemblies 210 of the multiple bed frame modules 200 from the storage configuration to the assembled configuration as described above and securing the first assemblies 210 to the central frame 26 to define the peripheral wall 22 of the modular bed frame assembly 20.

[0080] In certain embodiments, each of the bed frame modules 100, 200 can define a storage compartment when assembled delimited by the corresponding side panels. In some embodiments, the storage compartment can define a storage space shaped and dimensioned to store personal belongings such as sleeping accessories including, for example, pillows, sheets and covers.

[0081] While the above description provides examples of the embodiments, it will be appreciated that some features and/or functions of the described embodiments are susceptible to modification without departing from the spirit and principles of operation of the described embodiments. Accordingly, what has been described above has been intended to be illustrative and non-limiting and it will be understood by persons skilled in the art that other variants and modifications may be made without departing from the scope of the invention as defined in the claims appended hereto.

1. A modular bed frame assembly comprising:

a plurality of bed frame modules adapted to be assembled in an adjacent configuration to form the modular bed frame assembly, each of the bed frame modules defining at least a portion of a mattress receiving surface and comprising:

a first subassembly and a second subassembly configured to be assembled to form the respective bed frame module, each of the first and second subassemblies being configurable between an assembled configuration suitable for the assembly of the respective bed frame module, and a storage configuration.

2. The modular bed frame assembly of claim 1, wherein each of the first and second subassemblies comprises first and second panels, the first and second panels being adjustable between the storage and assembled configurations.

3. The modular bed frame assembly of claim 2, wherein, when the modular bed frame assembly is assembled, each of the first and second panels of the first subassembly defines a portion of a peripheral wall of the modular bed frame assembly.

4. The modular bed frame assembly of claim 3, wherein the first panel of the first subassembly comprises a slat-supporting rail.

5. The modular bed frame assembly of claim 2, wherein, when the modular bed frame assembly is assembled, each of the first and second panels of the second subassembly defines a portion of a central frame of the modular bed frame assembly.

6. The modular bed frame assembly of claim 5, wherein, when the modular bed frame assembly is assembled, the first

panel of the second subassembly defines a portion of a central support beam of the central frame.

7. The modular bed frame of claim 6, wherein the first panel of the second subassembly includes a slat-supporting rail.

8. The modular bed frame assembly of claim 2, wherein, when at least one of the first and second subassemblies is in the storage configuration, the respective first and second panels extend substantially parallel to one another.

9. The modular bed frame assembly of claim 8, wherein, when at least one of the first and second subassemblies is in the assembled configuration, the respective first and second panels extend at an angle to one another.

10. The modular bed frame assembly of claim 8, wherein, when at least one of the first and second subassemblies is in the assembled configuration, the respective first and second panels are disposed orthogonally to one another.

11. The modular bed frame assembly of claim 2, wherein the first panel is pivotably secured to the second panel, at least one of the first and second subassemblies including a locking hinge configured to retain the respective first panel at a given angle relative to the corresponding second panel when the corresponding one of the first and second subassemblies is in the assembled configuration.

12. The modular bed frame assembly of claim 1, wherein the first subassembly includes an upholstered external surface.

13. The modular bed frame assembly of claim 1, further comprising a modular headboard, the modular headboard including a plurality of headboard subassemblies, each of the plurality of headboard subassemblies being securable to a corresponding one of the plurality of bed frame modules.

14. The modular bed frame assembly of claim 13, wherein the modular headboard further includes a headboard cover sized to conceal the plurality of headboard subassemblies when the modular bed frame assembly is assembled.

15. The modular bed frame assembly of claim 1, wherein each of the plurality of bed frame modules is disposed sequentially with a first adjacent one of the bed frame modules along a transverse direction of the modular bed frame assembly and disposed sequentially with a second adjacent one of the plurality of bed frame modules along a longitudinal direction of the modular bed frame assembly.

16. A modular bed frame assembly comprising:

a plurality of central subassemblies adapted to be assembled to form a central frame of the modular bed frame assembly; and

a plurality of peripheral subassemblies, each of the peripheral subassemblies being configured to be secured to the central frame of the modular bed frame assembly to define at least a portion of a peripheral wall of the modular bed frame assembly;

wherein each of the central and peripheral subassemblies is configurable between an assembled configuration suitable for the assembly of the modular bed frame assembly, and a storage configuration.

17. The modular bed frame assembly of claim 16, wherein each of the central and peripheral subassemblies comprises first and second panels, the first and second panels being adjustable between the storage and assembled configurations.

18. The modular bed frame assembly of claim 17, wherein the first panel is pivotably secured to the second panel.

19. The modular bed frame assembly of claim 18, wherein when the corresponding one of the central and peripheral subassemblies is in the storage configuration, the first and second panels extend substantially parallel to one another.

20. The modular bed frame assembly of claim 19, wherein when the corresponding one of the central and peripheral subassemblies is in the assembled configuration, the first and second panels extend at an angle to one another.

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